
Curvature Partitioning in Multichannel Convolutional Dictionary Learning

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Abstract

Convolutional dictionary updates are usually distinguished by their solvers, although they differ in how they partition and approximate the same frequency-domain curvature. We formulate a common block quadratic construction that recovers equality-constrained ADMM with exact or conjugate-gradient inverses, consensus ADMM, and FISTA. It also distinguishes channels used as observation rows from channels used as dictionary right-hand sides. This construction yields a channel-contrast majorized consensus update that retains a low-rank channel subspace and bounds discarded curvature without changing the reconstruction objective. We prove its majorization and fixed-point consistency and derive full-iteration channel-coordinate equivalences. All four established methods reproduce independently executed SPORCO trajectories, with maximum relative objective error 6.92×10^{-11} . Both color representations stabilize on 40 RGB images at 25.54% and 32.80% of their initial objectives. The retained-curvature method reaches 32.94% and reduces local-copy storage, while exact grouped solves remain faster on RGB. Controlled correlation experiments identify when retaining channel curvature is useful and when its approximation becomes conservative.

1 Introduction

Convolutional dictionary learning (CDL) learns filters that can occur at any image location. Its alternating optimization has an asymmetry: sparse coding has a rank-one frequency-domain system for scalar signals, whereas updating the dictionary has a system whose rank can grow with the number of training images. Exact inversion, consensus splitting, and proximal gradients address this same curvature in different ways [Wohlberg, 2016a, Garcia-Cardona and Wohlberg, 2018b]. For multichannel signals, the situation also depends on the representation: channels can enlarge the observation dimension or the dictionary dimension.

The useful question is therefore not whether another solver can process color images. It is *which parts of the dictionary curvature must be solved exactly, which can be bounded, and which apparent differences are only changes of coordinates*. We answer it with a block quadratic construction that separates the representation, observation partition, local curvature, and enforcement of agreement. Its computational unit is a diagonal-plus-low-rank inverse, followed by the same support-and-normalization projection. Iterated Sherman–Morrison (ISM), conjugate-gradient (CG) updates, consensus ADMM, and constant-step FISTA follow by specifying these ingredients; their complete learning trajectories are checked against independently executed SPORCO solvers.

The construction also permits an intermediate choice: group the channels of an image, retain the curvature of their mean coefficient map, and bound the remaining contrast curvature. The resulting *channel-contrast majorized consensus* (CCMC) update retains the full residual gradient while using one rank-one inverse per image and frequency. It reduces the number of local dictionary copies relative to channelwise consensus without requiring a rank- C inverse. This combination, rather than the use of color data or consensus itself, is the algorithmic contribution.

We establish three structural results. A retained-subspace curvature completion majorizes each local Hessian and has the original constrained stationary equations at a fixed point. Orthogonal channel coordinates conjugate the entire learning iteration when channels belong to the dictionary atoms. The same claim fails when channels belong to independently penalized coefficient maps; those channels can instead be flattened into the image index. Experiments connect these distinctions to numerical agreement, convergence on natural color photographs, and the dependence of approximation quality on channel correlation. The study concerns optimization of the training functional, not downstream restoration quality or a global convergence guarantee.

2 Context and scope

Frequency-domain low-rank solves underpin efficient convolutional sparse coding [Wohlberg, 2014, 2016a]. Šorel and Šroubek [2016] developed inversion-lemma and consensus formulations, and Garcia-Cardona and Wohlberg [2018b] compared dictionary updates within a common alternation protocol, identified the relationship between volume-based and consensus updates, and introduced a competitive batch FISTA update. Auxiliary-variable coupling is consequential even when the two subproblem algorithms are unchanged [Garcia-Cardona and Wohlberg, 2017]; we retain that coupling throughout.

Multichannel atoms, shared scalar atoms, and inter-channel sparsity coupling are established representations [Wohlberg, 2016b]. The subsequent multichannel study of Garcia-Cardona and Wohlberg [2018a] treats product dictionaries, hyperspectral data, and proximal-gradient parameter selection. Our two color representations reproduce that established modeling capability. CCMC neither learns a spectral dictionary nor discards spectral components of the representation: its channel transform acts only inside a dictionary-update curvature model.

Majorization is also established in CDL. The BPG-M framework of Chun and Fessler [2018] constructs separable majorizers and studies acceleration and convergence; Zhang et al. [2021] use diagonal preconditioning in ADMM. Proximal-gradient consensus already has CDL implementations [Silva and Rodriguez, 2020], so a scalar local approximation or an averaged gradient is not a new method. Our specific addition is a channel-block consensus update with an explicit retained-subspace Gram matrix, an isotropic completion of its discarded curvature, and the corresponding full-gradient correction. We found no such instantiation in these works or the supplied SPORCO source. This is a bounded novelty claim about that combination, not a claim to have invented majorized ADMM. Distributed coordinate descent [Moreau and Gramfort, 2022] addresses a complementary spatial parallelization problem and is outside our fixed sparse-coding protocol.

3 A common quadratic construction

Representations and constraints. Let $s_{c,k}$ be channel c of training image k , with N spatial samples, M atoms, C channels and K images. We use circular convolution. Multichannel atoms (MC) share a coefficient map across channels:

$$F_{\text{MC}}(D, x) = \frac{1}{2} \sum_{c,k} \left\| \sum_m d_{c,m} * x_{m,k} - s_{c,k} \right\|_2^2 + \lambda \sum_{m,k} \|x_{m,k}\|_1. \quad (1)$$

Shared scalar filters (SF) instead have separate coefficient maps:

$$F_{\text{SF}}(d, x) = \frac{1}{2} \sum_{c,k} \left\| \sum_m d_m * x_{c,m,k} - s_{c,k} \right\|_2^2 + \lambda \sum_{c,m,k} \|x_{c,m,k}\|_1. \quad (2)$$

Each complete atom has unit Euclidean norm and prescribed spatial support. In MC its norm includes the channel dimension, $\sum_c \|d_{c,m}\|_2^2 = 1$. This is SPORCO’s executable Pcn convention. Equations (80) and (87) of Garcia-Cardona and Wohlberg [2018b] write the normalization with a per-channel index; implementing that literally would define a different feasible set from its color solver. We use the canonical executable convention consistently in every reference and proposed run. We do not impose a zero-mean constraint. If P restricts support, the atomwise projection is $\Pi_C(v) = Pv/\|Pv\|_2$ whenever $Pv \neq 0$. No zero projected atom occurs in the reported runs. The unit *sphere* is nonconvex,

so neither the dictionary subproblem nor the joint learning problem is globally convex under this constraint.

Fixing x , the spatial DFT reduces the fidelity to frequencywise matrix products $A_\omega d_\omega - s_\omega$. In MC, $A_\omega \in \mathbb{C}^{K \times M}$ is shared by C right-hand sides. In SF, $A_\omega \in \mathbb{C}^{CK \times M}$ has one right-hand side. Thus the upper bounds on dictionary-update rank are $\min(K, M)$ and $\min(CK, M)$ respectively. All unqualified sums and norms below include frequency and right-hand-side indices. The DFT can be unitary in the derivation; our code uses SPORCO’s standard real FFT normalization and evaluates the unscaled spatial functional.

Partition, model, and coupling. Partition the observation rows into B blocks A_b, s_b , and let $Eg = (g, \dots, g)$. The exact dictionary problem is

$$\min_{z, g} \sum_{b=1}^B f_b(z_b) + \iota_C(g), \quad z = Eg, \quad f_b(z) = \frac{1}{2} \|A_b z - s_b\|_F^2. \quad (3)$$

At an anchor a_b , replace f_b by its tangent quadratic

$$m_b(z; a_b) = f_b(a_b) + \text{Re}\langle \nabla f_b(a_b), z - a_b \rangle + \frac{1}{2} \|z - a_b\|_{Q_b}^2, \quad H_b = A_b^* A_b. \quad (4)$$

This is a majorizer if $Q_b \succeq H_b$ and is exact if $Q_b = H_b$. We parameterize $Q_b = V_b^* V_b + \tau_b I$, with τ_b allowed to vary with frequency. With a common consensus penalty $\sigma > 0$, minimizing this model plus the augmented quadratic gives the single local update

$$\underbrace{(V_b^* V_b + (\tau_b + \sigma) I)}_{Q_b + \sigma I} z_b^+ = A_b^* s_b + (Q_b - H_b) a_b + \sigma(g - u_b). \quad (5)$$

The remaining ADMM steps [Boyd et al., 2011] are independent of the local model:

$$\bar{z}_b = \alpha z_b^+ + (1 - \alpha)g, \quad g^+ = \Pi_C \left(B^{-1} \sum_b (\bar{z}_b + u_b) \right), \quad (6)$$

$$u_b^+ = u_b + \bar{z}_b - g^+. \quad (7)$$

We use $\alpha = 1$. In the majorized ADMM instances a_b is the previous unrelaxed local iterate. The code packs all blocks into array dimensions, constructs $(V_b, \tau_b, (Q_b - H_b)a_b)$, and executes equations (5)–(7) once. Different linear-solver kernels operate on that same system. They are reused from SPORCO, not redefined as complete dictionary algorithms.

Eliminated coupling and FISTA. There is a second way to enforce the equality in (3): eliminate the copies before forming the model. With $B = 1$, $u = 0$, $\sigma = 0$, and $Q = LI$, equation (5) followed by (6) is

$$g^{t+1} = \Pi_C(a^t - L^{-1} A^*(Aa^t - s)), \quad (8)$$

$$a^{t+1} = g^{t+1} + \frac{t_t - 1}{t_{t+1}}(g^{t+1} - g^t), \quad t_{t+1} = \frac{1 + \sqrt{1 + 4t_t^2}}{2}.$$

Here $t_0 = 1$; dual feedback is disabled. This is the FISTA update [Beck and Teboulle, 2009] used by SPORCO. It shares the quadratic solve/project driver, but is not asserted to be the zero-penalty limit of ADMM. Its fixed L gives a true majorizer only when $L \geq \|A\|_2^2$; the reference-matched heuristic does not certify this inequality throughout learning. In contrast, the completion constructed below certifies majorization at every local update. Convex FISTA rate theorems do not apply to our sphere-constrained, changing-coefficient problem.

Known updates and implementation parity. One exact block containing every observation gives the equality-constrained ADMM update. ISM and warm-started CG are alternative inverse kernels for it. Singleton exact blocks give rank-one consensus solves; parallel consensus changes their scheduling. Spatial tiling changes the spatial domain and boundary operator rather than only choosing an inverse kernel: exact reproduction requires preserving the independent-image convolution boundaries. We therefore analyze it as an operator embedding, not as a verified implementation. The copy-DFT relation to volume-based updates is derived in Appendix A.

Table 1: Instantiation by partition and local curvature. E: exact curvature; T: retained-subspace completion; S: scalar curvature. “Verified” denotes an implemented and measured instance; “algebraic” is a coordinate equivalence, not an independently timed solver.

Observation blocks	Exact + ADMM	T + ADMM	S + eliminated coupling
One global block	ISM, CG (verified)	Unmeasured	FISTA (verified)
Singleton rows	Consensus (verified)	Same if $q = 1$ (verified)	Assemble gradient first
Channels of each image	Grouped ISM (verified)	CCMC (verified)	Assemble gradient first
Copy DFT coordinates	Conjugate consensus (algebraic)	Conjugate model	Same global gradient

Initialization is part of the algorithm, not a nuisance parameter. For parity, we use the reference’s projected initial dictionary, $u^0 = g^0$ for the equality split, $u_b^0 = g^0/\sigma$ for consensus, and $a^0 = g^0$ for FISTA. CG starts its first frequency solution at zero and reuses subsequent solutions, with the reference’s relative tolerance and iteration cap. Only reference construction uses SPORCO’s solver dispatch; the template’s step method contains no dictionary-solver dispatch.

4 Retaining channel curvature

In SF, singleton consensus stores CK local copies, whereas grouping the C rows of each image needs K copies but ordinarily requires rank- C solves. The quadratic construction supplies intermediate choices. For each image and frequency, choose a fixed orthogonal channel matrix T and write

$$TA_b = \begin{bmatrix} V_b \\ R_b \end{bmatrix}, \quad V_b \in \mathbb{C}^{q \times M}, \quad \tau_b = \|R_b\|_F^2, \quad Q_b = V_b^* V_b + \tau_b I. \quad (9)$$

Our T is the real orthonormal DCT-II, whose first row is $C^{-1/2}(1, \dots, 1)$. The primary instance keeps $q = 1$. A channel mean is inexpensive and appropriate when coefficient maps are positively correlated; it is not assumed to be a principal component. The right side of (5) becomes

$$A_b^* s_b + (\tau_b I - R_b^* R_b) a_b + \sigma(g - u_b). \quad (10)$$

In particular, all channels still contribute to the gradient. Applying a low-rank approximation directly to $A_b^* s_b$ would solve a different problem.

Proposition 1 (Majorization and consistency). *The model (9) satisfies $Q_b \succeq H_b$. Its local minimizer is unique for $\sigma > 0$. At a feasible fixed point of (5)–(7), with unchanged coefficients and $a_b = z_b = g$, the first-order conditions are those of (3).*

Proof. Orthogonality gives $H_b = V_b^* V_b + R_b^* R_b$. For every vector v ,

$$v^*(Q_b - H_b)v = \|R_b\|_F^2 \|v\|_2^2 - \|R_b v\|_2^2 \geq 0. \quad (11)$$

Thus $Q_b + \sigma I \succ 0$. At a fixed point, the model correction cancels against $Q_b g$, yielding $\nabla f_b(g) + \sigma u_b = 0$. Optimality of the support-and-sphere projection gives $\sigma \sum_b u_b \in N_C(g)$, with N_C the normal space of the supported product of spheres. Summing the local equations gives $0 \in \sum_b \nabla f_b(g) + N_C(g)$. $\square$

This result establishes local model validity and fixed-point consistency. It does not prove convergence of the alternating, nonconvex learning sequence. We assess stabilization empirically and report dictionary changes and consensus residuals as well as objective values.

Equation (5) is solved using the same inverse kernels as before. For $q = 1$, writing $V_b = v_b^*$ and $\eta = \sigma + \tau_b$, the inverse is

$$(\eta I + v_b v_b^*)^{-1} r = \eta^{-1} \left(r - \frac{v_b (v_b^* r)}{\eta + v_b^* v_b} \right). \quad (12)$$

Setting $q = 0$ gives a scalar majorized ADMM update, included as an ablation. Retaining every row gives grouped exact consensus. Setting the block size to one and retaining its row gives ordinary consensus with *the same driver and model builder*; this provides a numerical reduction to an external reference. These endpoints also delimit novelty: neither grouping alone nor scalar majorization is the proposed contribution.

Mechanism and computational cost. For an SF channel block, forming gradients and the discarded-curvature correction costs $O(CM)$ per frequency after the channel transform. A general dense channel transform costs $O(C^2M)$ in our implementation; a fast DCT could reduce this for large C . Rank- q ISM costs $O(q^2M)$ instead of $O(C^2M)$ for the inverse. The K local copies cost $O(KMN)$ versus $O(CKMN)$ for singleton consensus, although coefficient storage remains $O(CKMN)$ for either method. Total memory therefore need not decrease by a factor C . The largest eigenvalue of the discarded Gram matrix is bounded by its trace; when discarded energy is large or dispersed across channels, the trace bound can be conservative. There is no uniform speed advantage over exact grouped solves.

5 Which changes of coordinates preserve the algorithm?

Proposition 2 (Channel equivariance). *For MC, let $U \in \mathbb{R}^{C \times C}$ be orthogonal and transform $S \mapsto US$, $D^0 \mapsto UD^0$. With identical scalar parameters and transformed dictionary states, exact ADMM, consensus, and scalar-curvature FISTA produce $D^t \mapsto UD^t$ and unchanged coefficient maps at every iteration. CG has the same property in exact arithmetic with its global relative-residual stopping rule.*

Proof. The sparse-coding normal matrix is $\sum_c D_c^* D_c$, and its data term is $\sum_c D_c^* S_c$; both are invariant under U . Hence its primal, thresholded, and dual coefficient states remain unchanged. In the dictionary update, A_b is independent of the output channel, so its inverse and gradient commute with U . Support restriction commutes with U , and each complete atom norm is invariant: $\Pi_C(Uv) = U\Pi_C(v)$. Averaging, dual feedback, and scalar extrapolation commute as well, completing induction. The block CG operator commutes with U and preserves the residual norm, so its recurrence and stopping decision are conjugate. $\square$

An RGB/opponent-color change of basis thus does not define a new MC algorithm. This conclusion depends on complete-atom normalization and common channel metrics; independent channel normalizations would break the projection argument.

For SF, the equality $j = (c, k)$ instead gives an exact flattening to scalar-image CDL with CK observations. With the same row order, initial states and penalty, every step of the ungrouped algorithm is identical. An arbitrary orthogonal rotation of SF coefficient channels preserves the squared residual but generally changes $\sum_c \|x_c\|_1$. It is therefore not an equivalent representation. A joint $\ell_{2,1}$ penalty is an established alternative sparse-coding model, not a new dictionary update [Wohlberg, 2016b, Garcia-Cardona and Wohlberg, 2018a]. These distinctions explain why MC channels are interchangeable right-hand-side coordinates, whereas SF channel grouping is a meaningful algorithmic decision.

6 Numerical evidence

Protocol. We use MIRFLICKR-derived photographs [Huiskes et al., 2010], center cropping, resizing, scaling to $[0, 1]$, and subtraction of a Tikhonov lowpass component with parameter 5. The image selection and all initial dictionaries are seeded and logged. All runs use $\lambda = 0.1$, support 8×8 , unit complete-atom norms, and one auxiliary-coupled sparse-coding/dictionary iteration per outer step. The two color representations use 40 RGB images of 256×256 pixels and 64 atoms. We execute a matched SPORCO [Wohlberg, 2017] ConvBPDNDictLearn reference for each representation at the same regularization, rather than importing settings from its color examples. Smaller grayscale comparisons use four 48×48 images and 16 atoms; additional color tests use eight 64×64 images and 32 atoms.

We call an endpoint *stabilized* when, after at least 500 iterations, the range of the objective over the most recent 50 iterations is at most 10^{-3} of its latest value. The complete stopping and logging protocol is in Appendix B. Reported functionals are computed after both updates, using projected filters, not an intermediate ADMM variable. For agreement we measure

$$\epsilon_F = \max_{t \in \mathcal{T}} \frac{|F_t - F_t^{\text{ref}}|}{|F_t^{\text{ref}}|}, \quad \epsilon_D = \frac{\|D_T - D_T^{\text{ref}}\|_F}{\|D_T^{\text{ref}}\|_F}, \quad (13)$$

without filter permutation or sign alignment. Grayscale comparison logs include every iteration. Identical settings include initialization, penalties, relaxation, CG stopping, FISTA momentum, and

Table 2: Latest stabilized objectives and numerical agreement. Gray: $K = 4$, 48^2 pixels, $M = 16$, mean $\pm$ sample standard deviation across two initializations. Color: $K = 40$, 256^2 pixels, $C = 3$, $M = 64$, one initialization. Settings are (B, r, Q) : number of blocks, rows per block, and E/T/S curvature. All dictionary penalties and steps are in Appendix B. ϵ_F is the maximum relative trajectory error; ϵ_D is the final relative dictionary error (worst seed). A dash marks a run without a paired reference. CCMC[†] reports errors of its singleton reduction on RGB data, not a reference for its nondegenerate update.

Method	Rep.	(B, r, Q)	ϵ_F	ϵ_D	Functional
ISM	Gray	(1, 4, E)	5.08×10^{-16}	1.08×10^{-14}	20.581 ± 0.308
CG	Gray	(1, 4, E)	6.92×10^{-11}	2.99×10^{-9}	20.690 ± 0.273
Consensus	Gray	(4, 1, E)	8.61×10^{-15}	9.18×10^{-13}	20.605 ± 0.041
FISTA	Gray	(1, 4, S)	8.65×10^{-16}	9.22×10^{-14}	20.552 ± 0.087
FISTA	MC	(1, 40, S)	4.21×10^{-8}	5.89×10^{-6}	5584.12
FISTA	SF	(1, 120, S)	8.08×10^{-8}	7.83×10^{-6}	7172.75
Consensus	SF	(120, 1, E)	—	—	7181.90
Grouped exact	SF	(40, 3, E)	—	—	7180.87
Scalar MM	SF	(40, 3, T ₀)	—	—	7289.26
CCMC [†]	SF	(40, 3, T ₁)	2.86×10^{-16}	1.96×10^{-15}	7202.52

the absence of backtracking. The plan and its timestamp precede validation. All attempted numerical configurations are retained; no failed convergence runs are removed from the tables.

Agreement of complete learning trajectories. Across the two initializations, all four methods agree with SPORCO: the largest trajectory deviation is 6.92×10^{-11} and the largest final dictionary error is 2.99×10^{-9} . The exact ISM and consensus methods have worst errors 8.61×10^{-15} and 9.18×10^{-13} , respectively for the two metrics. The larger, but still very small, CG discrepancy is consistent with an inexact solve and finite-precision stopping. Figure 1 overlays complete trajectories for one initialization; both are tabulated in the appendix. The original 300-step comparisons are also retained, but their objectives had not yet satisfied the stabilization criterion, motivating the explicitly logged extensions.

Color representations. Both full-size color runs stabilize at 500 iterations. MC reaches 5584.12 (25.54% of its zero-code initial value), and SF reaches 7172.75 (32.80%). Their maximum relative deviations from their own matched references are 4.21×10^{-8} and 8.08×10^{-8} (Figure 2). These are different representations with different coefficient penalties and atom dimensions; their raw objective values are not a controlled ranking of representational quality. They demonstrate that the same dictionary-update construction handles both layouts without changing sparse coding or rescaling the objective. The maximum atom-norm error across stabilized learning runs is 1.79×10^{-7} .

The full-size computations use float32 on one GPU. Before running them, a separate float64 CPU/GPU audit of actual SPORCO source execution found a maximum dictionary difference of 7.22×10^{-15} across grayscale and both color representations. This checks the backend adapter independently of the proposed framework; CPU-only agreement is the primary validation for all four update families.

The retained-curvature update. On full-size SF color data, CCMC stabilizes after 610 iterations at 7202.52, or 32.94% of its initial objective. Grouped exact consensus reaches 7180.87 and channelwise consensus reaches 7181.90; CCMC’s endpoint is 0.30% above grouped exact. The scalar-curvature ablation needs 1010 iterations and stabilizes at 7289.26 (Figure 3). Keeping the channel mean therefore contributes beyond merely reducing the number of copies. CCMC’s discarded coefficient energy is 18.3% at its endpoint. All four methods also stabilize for both smaller-color initializations; the complete ablation table and curves are in the appendix.

Setting CCMC’s blocks to single observation rows while retaining their one row reproduces the independently executed RGB SPORCO consensus reference, with $\epsilon_F = 2.86 \times 10^{-16}$ and $\epsilon_D = 1.96 \times 10^{-15}$. The grayscale reduction passes as well. A separate check on 48 frequency/block systems from real-image coefficients compares the retained inverse to a dense solve, obtaining maximum relative error 2.80×10^{-16} ; the smallest eigenvalue of $Q - H$ is -3.20×10^{-15} , consistent with

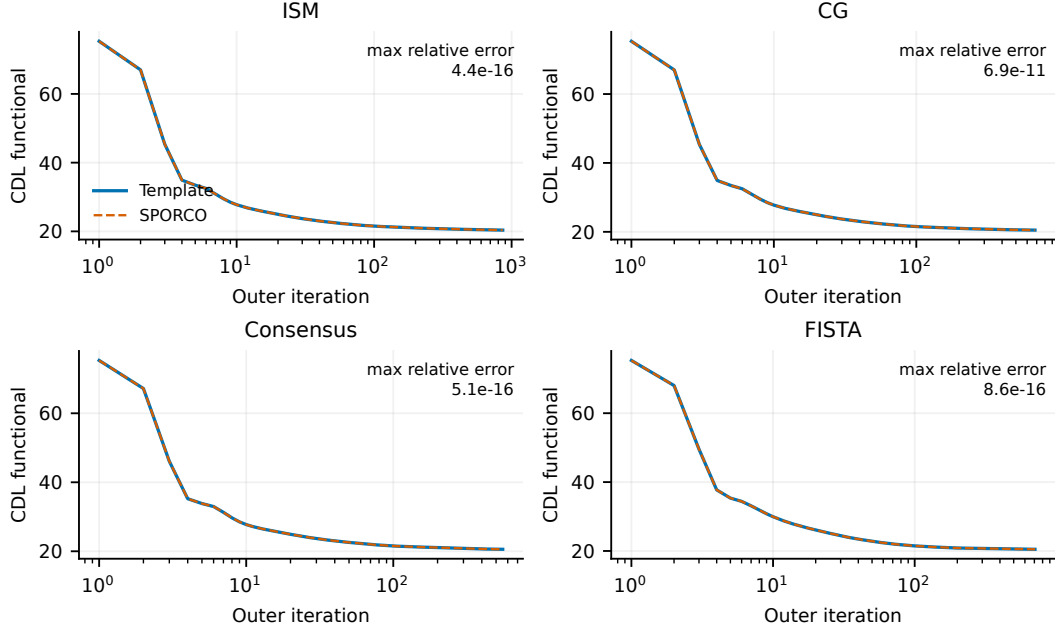


Figure 1: Template and SPORCO learning trajectories from the same grayscale inputs and initialization. Curves overlap; errors are measured at every iteration, not inferred from the plots. The horizontal axis is logarithmic to show the early transient and the stabilized endpoint.

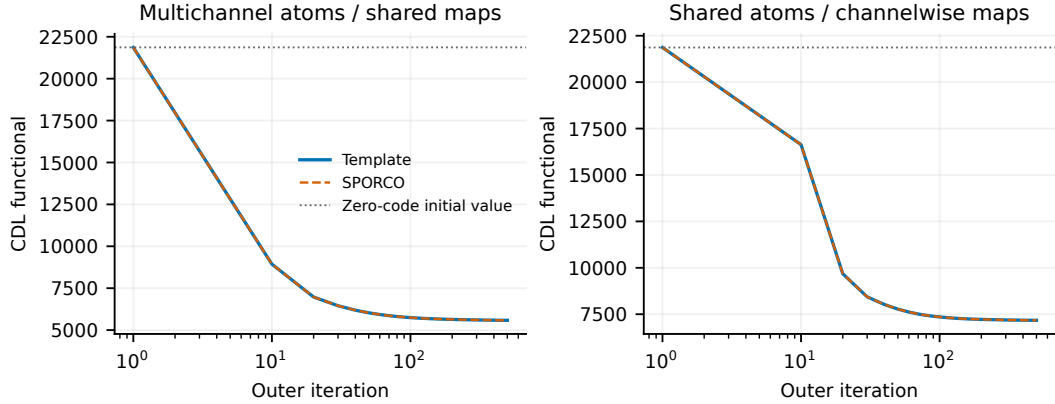


Figure 2: Both multichannel representations on 40 RGB photographs, with 64 atoms and $\lambda = 0.1$. Each template is overlaid on its own matched SPORCO reference. The dotted level is the actual objective at zero coefficients.

roundoff at zero. These checks distinguish the proposed quadratic construction from replacing a named solver with an unrelated implementation.

Equivalence and its boundary. The orthogonal-channel experiment applies the same random channel rotation to MC inputs and initial atoms and compares 100 complete iterations, including sparse coding. The largest conjugated dictionary error across all four methods is 8.61×10^{-15} . SF channel/image flattening gives identically equal dictionary sequences in the tested ISM, consensus and FISTA runs. As a negative control, rotating SF input and coefficient channels preserves its fidelity to roundoff but changes its ℓ_1 penalty by 4.40%. Thus the algebra predicts both an equivalence and a concrete failure of an apparently similar transformation.

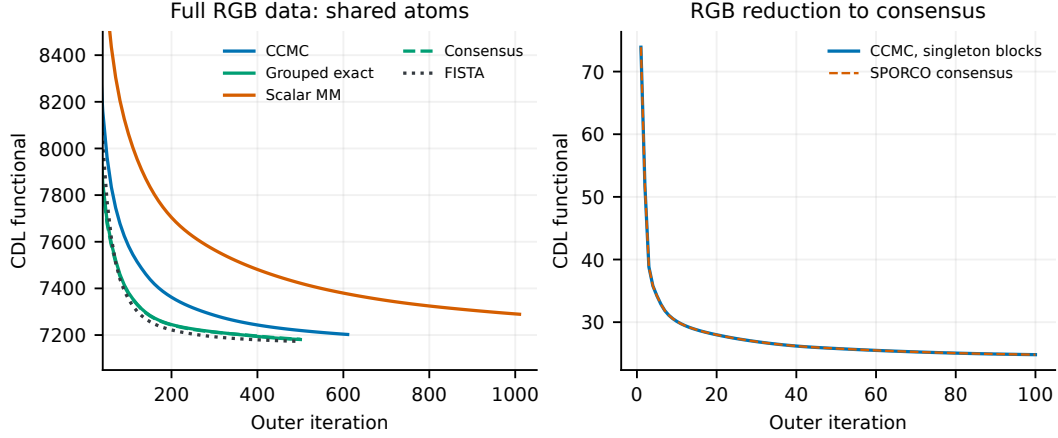


Figure 3: Left: full-size shared-filter learning with retained, exact and scalar local curvature, plus consensus and FISTA. The expanded scale displays iterations after the initial decrease; complete curves are logged. Right: reducing the proposed curvature construction to singleton blocks matches SPORCO consensus on RGB images.

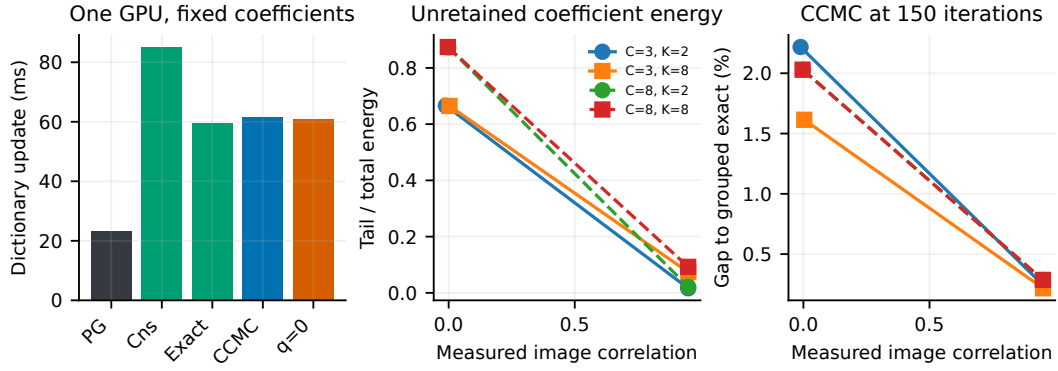


Figure 4: Measured cost and approximation mechanism. Left: synchronized full-size dictionary-update timings on one H200 GPU at common coefficients (PG: FISTA). Center/right: all shared-filter correlation configurations, showing coefficient-tail energy and the CCMC objective gap relative to grouped exact at 150 iterations. The correlation study uses controlled transformations of natural photographs.

Correlation and cost. The controlled image transformations vary $C \in \{3, 8\}$, $K \in \{2, 8\}$ and channel correlation, keeping support, initialization rule, and 150-iteration budgets fixed. All 56 method/configuration endpoints are reported in the appendix. For highly correlated channels, CCMC discards only 1.6–9.3% of coefficient energy and ends 0.22–0.29% above grouped exact. For weakly correlated channels it discards 66.5–87.4% and the gap grows to 1.61–2.22% (Figure 4). In the correlated eight-channel cases, measured CPU dictionary-update time is 32.9–35.0% lower than grouped exact at the same iteration count. These small controlled experiments support the curvature mechanism; they do not establish performance on hyperspectral sensor data.

A separate full-size GPU microbenchmark uses common fixed coefficients. Median dictionary-update times are 23.2 ms for FISTA, 84.9 ms for channelwise consensus, 59.5 ms for grouped exact, and 61.6 ms for CCMC. Thus CCMC saves 27.5% relative to channelwise consensus in that benchmark, but is slightly slower than grouped exact and substantially slower than FISTA. Its dual array occupies 640 MiB versus 1920 MiB for channelwise consensus. These are copy-storage measurements, not total solver memory. The six full-size learning runs, including paired references for the two FISTA representations, consumed 545.7 seconds of measured iterative execution; the stabilized grayscale and smaller-color learning runs used 995.3 seconds summed over the CPU iteration timings. Setup, file I/O and the separately reported diagnostics are additional costs.

7 Implications and limitations

The structural choice precedes the solver choice. In MC, increasing C raises the sparse-coding rank while the dictionary update retains rank at most K with C right-hand sides. In SF, sparse coding remains rank one per channel/image, but the dictionary update can have rank CK . This is the channel/image rank duality emphasized by Garcia-Cardona and Wohlberg [2018b,a], now expressed in the same observation operator. For small observation rank, an exact inverse avoids surrogate looseness. For many observations, singleton consensus exposes independent rank-one systems at the cost of copies, and global FISTA avoids inversions and local dual storage. With many correlated SF channels, grouping plus retained curvature offers an intermediate computational point. Weakly correlated channels favor retaining more directions, exact grouped solves, or a global gradient update; channel count alone is insufficient to choose.

Several limitations are material. First, the reported criterion measures objective stabilization, not convergence to a stationary point. At the full CCMC endpoint the relative dictionary change is 9.11×10^{-4} and the consensus residual is 7.23×10^{-3} , both nonzero. Second, a fixed channel-mean subspace can miss anticorrelated or spectrally localized structure, and our dense DCT implementation has quadratic channel-transform cost. Third, we have two initializations at small scale and one at full scale, with no held-out restoration task or parameter search. Timings characterize these implementations and hardware, not universal algorithm rankings. Fourth, MC complete-atom normalization follows the reference code and differs from a literal per-channel reading of the earlier equations. Finally, spatial masks break the simple frequencywise data operator unless additional splitting is introduced; masked and joint-sparsity methods already exist in SPORCO and are not implemented extensions here. These restrictions, including the unfavorable RGB timing comparison and weaker low-correlation results, are part of the evidence rather than omitted exceptions.

8 Conclusion

Convolutional dictionary updates can be organized around one local quadratic solve and one global constraint projection. Observation partitioning determines local rank and copy storage; curvature selection determines how much of that rank is inverted; the representation determines whether channels are observations or right-hand sides. The construction reproduces four established learning algorithms, gives a concrete retained-channel-curvature method, and exposes both coordinate equivalences and their limits. The resulting guidance is conditional: retain inexpensive correlated curvature when it is informative, and use the exact or gradient endpoints when its discarded energy or implementation cost dominates.

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A Copy coordinates and the volume construction

Consider equal-sized observation blocks and write $\mathcal{A} = \text{diag}(A_1, \dots, A_B)$, $s = (s_1, \dots, s_B)$, and $E = \mathbf{1}_B \otimes I$. Let W be the unitary B -point DFT and transform the copy and residual coordinates together:

$$\tilde{z} = (W^* \otimes I)z, \quad \tilde{s} = (W^* \otimes I)s, \quad \tilde{\mathcal{A}} = (W^* \otimes I)\mathcal{A}(W \otimes I), \quad \tilde{E} = \sqrt{B}e_0 \otimes I. \quad (14)$$

The left and right identity factors have the appropriate observation and atom dimensions. The transformed problem is

$$\min_{\tilde{z}, g} \frac{1}{2} \|\tilde{\mathcal{A}}\tilde{z} - \tilde{s}\|_F^2 + \iota_C(g), \quad \tilde{z} = \tilde{E}g. \quad (15)$$

The transformed operator is block circulant: its (ℓ, j) block is $B^{-1} \sum_b \exp(2\pi i b(\ell - j)/B) A_b$. Thus convolution in a copy dimension and a constraint supported on its first slice are coordinate descriptions of the consensus problem. Both the fidelity and the augmented norm are invariant, so transformed primal and dual initial states yield transformed ADMM iterates. The global update remains

$$g^+ = \Pi_C \left(B^{-1} \tilde{E}^* (\tilde{z}^+ + \tilde{u}) \right), \quad \tilde{E}^* \tilde{E} = BI. \quad (16)$$

For MC, tensoring these transformations with I_C leaves the proof unchanged. This extends the consensus/volume relationship discussed by [Garcia-Cardona and Wohlberg \[2018b\]](#) while making the normalization explicit. With an unnormalized copy DFT, the objective and penalty scales must be transformed consistently. Applying a DFT to the observation rows alone while keeping independent local dictionaries and unchanged penalties does *not* establish trajectory equivalence. We did not implement or time a separate volume solver.

B Experimental and implementation details

Data. We sample without replacement from the lexicographically sorted JPEG members of the supplied MIRFLICKR-25k archive, using NumPy’s default random generator with seed 2026. Each image is cropped to its central maximal square and resized with Pillow’s Lanczos filter. Integer RGB samples are divided by 255. Grayscale conversion, when used, is Pillow’s RGB-to-L conversion after resizing. The lowpass component is computed by SPORCO’s `tikhonov_filter` with parameter 5 and padding 16 and then subtracted. These are MIRFLICKR-derived data with the original preprocessing pattern, not the exact private MIRFLICKR-1M image selection in the earlier review. The archive member names, crop coordinates, input hashes, array hashes and initial objectives are in `data/manifest.json`.

The 48×48 agreement experiments use the first four selected photographs, with independent random dictionary seeds 2026 and 2027. The 64×64 color experiments use the first eight photographs and both dictionary seeds. The full color experiment uses all 40 photographs and dictionary seed 2026. All random dictionaries are projected by the same SPORCO support-and-sphere operator before learning. The smaller experiments use 16 and 32 atoms respectively; the initial smoke tests use eight atoms. Every atom has 8×8 spatial support.

Alternation and stopping. Each outer step invokes one SPORCO ADMM CBPDN iteration at $\lambda = 0.1$ and $\rho_x = 3$, transfers its thresholded auxiliary coefficient maps, executes one dictionary update, and transfers the projected dictionary. Automatic penalty adaptation and relaxation are disabled. ISM/CG use $\sigma = 0.5J + 7$, where J is the number of observation rows; singleton consensus uses $\sigma = 2.2$; channel-block methods use $\sigma = 2.2C$. The latter keeps the aggregate consensus penalty $B\sigma$ equal to channelwise consensus. CG uses relative residual tolerance 10^{-3} , at most 1000 inner iterations, and the preceding unrelaxed frequency solution as its warm start. The SPORCO CG function returns a solver status code rather than an actual iteration count; we do not mislabel it as a measured count.

FISTA has no backtracking or monotone restart. Its Nesterov state persists across outer iterations. We use $L = 14K$ for scalar images and $L = 14CK$ for both color representations, matching the predeclared reference settings. The MC color multiplier follows the stability adjustment in the supplementary experiment of [Garcia-Cardona and Wohlberg \[2018b\]](#); the SF multiplier reflects its CK observation rows.

All objective values use the latest projected dictionary and auxiliary coefficients, after both substeps, with a direct spatial squared-residual evaluation and the unscaled ℓ_1 term. We never use a stale sparse-coding objective as a learning objective. Initial objectives use zero coefficients. Full-size float32 runs accumulate the objective terms in float64. Stabilization means at least 500 operators and a relative range at most 10^{-3} over the most recent 50 iterations, evaluated at the logged points. Agreement runs log every iteration; full color runs log every ten iterations; smaller color runs log every five. We also save dictionary changes and consensus residuals. This is an operational objective criterion, not a stationarity certificate. The first 300-iteration agreement runs are retained separately, and their subsequent longer runs were declared in a timestamped plan addendum.

Backend validation. The supplied SPORCO checkout is version 0.2.3.dev0. Its CuPy mirror contains sparse coders and base optimizers but no batch dictionary-learning module. We extend its own source-module

patching mechanism to execute the supplied `ccmod.py` and `cbpdnd1.py` sources with the existing CuPy primitives. The only host compatibility handling is NumPy’s removed `product` alias and shape-tuple products, which CuPy’s reduction rejects. No reference update equation is edited. Before the full runs, we compare CPU SPORCO, GPU SPORCO, and the independent template for ISM, consensus and FISTA on grayscale and both color representations for 30 iterations in float64. The complete arrays of errors are retained in `logs/backend_validation/`. CG agreement uses the unmodified CPU solver. Full color computations use only GPU device 0; the observed device is an NVIDIA H200, rather than the H100 listed in the resource description. CPU experiments use one FFT/BLAS thread per process.

Controlled correlation experiment. The mechanism study subsamples the 64×64 highpass photographs to 32×32 and averages their RGB channels. Each source is demeaned and scaled to RMS 0.12. Synthetic channel c of scene k mixes the scene with a deterministically shifted other photograph as $\sqrt{a} s_k + \sqrt{1-a} r_{c,k}$, followed by per-channel RMS normalization to 0.12. We use $a \in \{0, 0.95\}$, $C \in \{3, 8\}$, $K \in \{2, 8\}$, 16 atoms, dictionary seed 818 and 150 iterations. Actual empirical correlations are reported; the mixing coefficient is not assumed to equal the measured correlation. These are controlled image transformations, not collected hyperspectral measurements. There are seven methods per configuration: five SF updates and MC consensus/FISTA. No full-size convergence claim is based on this diagnostic.

Timing. We report descriptive end-to-end learning times, and separately benchmark dictionary updates at a common full-size SF dictionary and coefficient set. The latter uses five warmup updates followed by 30 GPU-synchronized timings, includes the coefficient FFT, projection and template diagnostics, and excludes sparse coding. It runs after other GPU experiments have finished. Timings of reference objects include SPORCO’s bookkeeping and are not treated as evidence of a speedup by the template. Local-copy storage is calculated from actual array byte counts; allocator-pool sizes are separately logged and are not presented as minimal algorithm memory.

C Additional numerical results

Table 3: Every stabilized grayscale run. The earlier 300-step and 30-step diagnostic runs remain in the artifact; their objective deviations are summarized below.

Method	Seed	Iterations	Functional	ϵ_F	ϵ_D
ISM	2026	862	20.363829	4.37×10^{-16}	1.02×10^{-14}
ISM	2027	593	20.798832	5.08×10^{-16}	1.08×10^{-14}
CG	2026	681	20.497018	6.92×10^{-11}	2.99×10^{-9}
CG	2027	500	20.883281	4.60×10^{-13}	1.77×10^{-11}
Consensus	2026	555	20.576165	5.06×10^{-16}	1.25×10^{-14}
Consensus	2027	677	20.634350	8.61×10^{-15}	9.18×10^{-13}
FISTA	2026	707	20.490272	8.65×10^{-16}	7.01×10^{-14}
FISTA	2027	500	20.613439	5.12×10^{-16}	9.22×10^{-14}

The maximum objective errors in the original 300-step runs and 30-step smoke experiments were 3.71×10^{-15} and 3.95×10^{-16} , respectively. They are numerical agreement checks, not stabilized endpoints.

Table 4: All smaller color runs: SF, $K = 8$, 64^2 pixels, $C = 3$, $M = 32$. Each row is a completed objective-stabilized run.

Method	Seed	Iterations	Functional	F/F_0	$\ \Delta D\ /\ D\ $	Primal residual
CCMC	2026	505	179.4488	0.2915	7.52×10^{-4}	9.69×10^{-4}
CCMC	2027	500	179.7215	0.2919	7.79×10^{-4}	9.84×10^{-4}
Grouped exact	2026	520	179.2391	0.2912	8.77×10^{-4}	1.08×10^{-3}
Grouped exact	2027	610	178.8081	0.2905	8.24×10^{-4}	9.89×10^{-4}
Scalar MM	2026	970	180.5490	0.2933	5.39×10^{-4}	6.98×10^{-4}
Scalar MM	2027	850	180.6621	0.2935	5.42×10^{-4}	6.32×10^{-4}
Consensus	2026	530	179.0936	0.2909	8.19×10^{-4}	1.20×10^{-3}
Consensus	2027	550	179.0668	0.2909	9.19×10^{-4}	1.26×10^{-3}

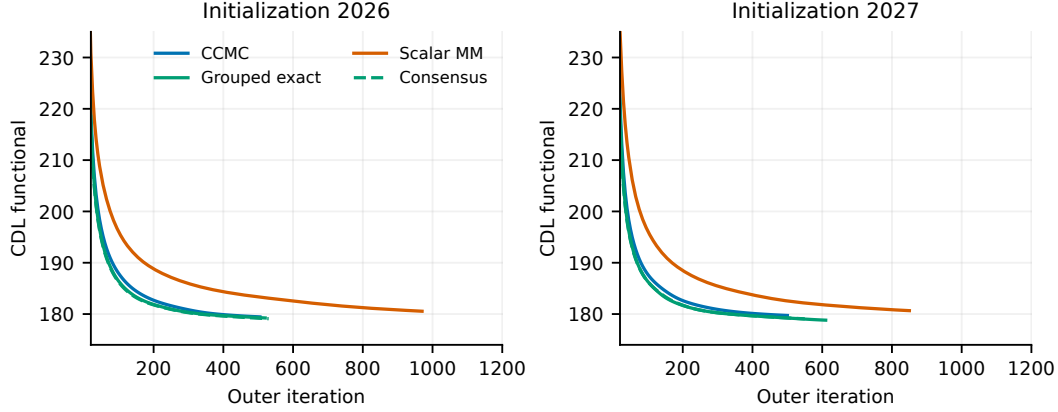


Figure 5: Both initializations and all smaller-color ablations. The vertical scale is expanded after the initial decrease; complete trajectories are stored.

Table 5: Full color stopping diagnostics. No row is interpreted as a certificate of stationarity. Times are measured template iteration totals, excluding separate objective logging and reference execution.

Method	Rep.	Iter.	Window range	$\ \Delta D\ /\ D\ $	Primal residual	Time (s)
FISTA	MC	500	5.72×10^{-4}	1.11×10^{-3}	—	27.29
FISTA	SF	500	4.17×10^{-4}	5.77×10^{-4}	—	46.62
CCMC	SF	610	9.74×10^{-4}	9.11×10^{-4}	7.23×10^{-3}	82.39
Grouped exact	SF	500	7.51×10^{-4}	8.20×10^{-4}	6.96×10^{-3}	64.34
Scalar MM	SF	1010	9.91×10^{-4}	5.56×10^{-4}	2.52×10^{-2}	131.66
Consensus	SF	500	8.31×10^{-4}	1.03×10^{-3}	1.02×10^{-2}	77.29

Table 6: All 150-iteration mechanism endpoints. SF columns are consensus (Cns), grouped exact (GE), CCMC, scalar MM ($q = 0$), and FISTA (PG); MC columns are consensus and FISTA. Correlations are measured, not nominal.

C, K	Corr.	SF Cns	SF GE	SF CCMC	SF $q = 0$	SF PG	MC Cns	MC PG
3, 2	-0.011	16.67	16.68	17.05	17.20	16.33	15.81	16.22
3, 2	0.950	12.24	12.24	12.27	13.14	11.65	8.85	9.31
3, 8	0.003	71.27	71.32	72.47	72.93	70.49	83.24	84.67
3, 8	0.950	73.73	73.74	73.90	75.30	72.89	52.09	52.38
8, 2	-0.001	47.51	47.55	48.51	48.62	47.00	44.95	46.66
8, 2	0.950	31.96	31.97	32.06	34.70	31.16	17.55	18.35
8, 8	-0.003	190.05	190.19	194.05	194.50	187.98	260.42	264.82
8, 8	0.950	196.14	196.17	196.73	200.62	194.42	99.60	102.09

D Reproducibility and scope of the artifact

All numbers and graphics in this document are generated from recorded runs by `code/analyze.py`. The source contains a single `DictionaryTemplate.step`, with representation packing, curvature construction and numerical inverse kernels as separate ingredients. `TemplateCDL` uses SPORCO’s unchanged sparse coder. Reference objects are created only by the comparison harness. `code/framework_plan.md` and its timestamped JSON digest precede validation; later protocol refinements are appended to `framework_plan_addenda.jsonl`. `codebase_notes.md` lists inspected and reused modules, and `attempts_log.md` records development failures and their resolution. Input and source hashes accompany each learning run. Final dictionaries, atom norms, trajectories, reference statistics, and full-size anchor coefficients are saved. The supplied `neurips.sty` is copied without modification.